

Bishnu Prasad Sahu

Bhilai, India | +91 9301105706 | mebishnusahu@gmail.com | linkedin.com/in/mebishnusahu05 | github.com/mebishnusahu0595

PROFESSIONAL SUMMARY

Full Stack Engineer and AI/ML specialist with 2+ years of hands-on production experience across Python, Computer Vision, distributed systems, and REST API development. Shipped 10+ production-grade applications spanning real-time inference systems, open-source SDKs, and agentic AI workflows. Contributed to Supabase and AOSSIE, won 2 competitive hackathons, and published peer-reviewed research. Proven track record of delivering measurable business outcomes — 40% API latency reduction, 95%+ tracking accuracy, and 80%+ error reduction across client-facing systems. Seeking to apply deep technical expertise in ML and distributed systems to large-scale production AI challenges.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, C++, Go, C, SQL

Frameworks & Libraries: Node.js, Express.js, React (Vite), Flask, .NET (C#), TensorFlow, OpenCV, YOLOv8, Deep SORT

AI / Machine Learning: Computer Vision, Deep Learning, Object Detection & Tracking, LLM Integration, Agentic AI, Multi-Agent Systems, Graph Neural Networks, NLP

Databases: MySQL, MongoDB, SQL Server, Supabase (PostgreSQL)

DevOps & Tools: Git, GitHub Actions, Docker, CI/CD Pipelines, CMake, REST APIs, Power BI

Computer Science Concepts: Real-time Inference, Distributed Systems, Data Pipelines, Red-Black Trees, Topological Sorting, Graph Algorithms, Dependency Resolution, AsyncLocalStorage

Other: Open-Source Contribution, Technical Writing, Peer-Reviewed Research, Hackathon Engineering, Agile Development

PROFESSIONAL EXPERIENCE

Full Stack Developer

Apr 2025 – Present

Oncorg

Remote

- Delivered 3+ full-stack web applications in C# / .NET within an agile team, reducing average feature delivery time by **~30%** through architecting reusable REST API components and standardized module templates.
- Optimized complex SQL Server queries for high-frequency data operations, achieving **40% reduction** in average API response time across core production endpoints.
- Collaborated cross-functionally with designers and product managers to translate business requirements into scalable, maintainable backend services.
- Implemented robust error handling, logging, and monitoring practices that improved system observability and reduced incident response time.

Open Source Contributor

Jan 2026 – Present

Supabase

Remote

- Identified and resolved a critical storage URL initialization bug in **supabase-py** that caused runtime failures across **100% of storage SDK calls**, restoring full client-side storage functionality.
- Authored 12+ unit tests for Sync and Async clients covering 5 base URL edge-case scenarios, raising storage module test coverage from **0% to a measurable baseline** and enforcing regression safety.
- Engaged with maintainers via pull request reviews and issue triaging, contributing to code quality and open-source project velocity.

Full Stack Developer

Jan 2026 – Feb 2026

Code2Dbug

Bhilai, India

- Built and deployed **4 full-stack web applications** using Node.js, Express.js, and React (Vite) within a 6-week engagement with 100% on-time delivery.
- Diagnosed and resolved critical frontend-backend integration failures across 2 legacy projects, reducing user-facing errors by **over 80%** and improving application reliability.
- Established consistent API contracts and component architecture patterns to improve long-term code maintainability for the engineering team.

AI / Computer Vision Developer

Sep 2024 – Nov 2025

CogniAble

Remote

- Engineered a production real-time person detection and tracking system using **YOLOv8 + Deep SORT**, achieving **95%+ tracking accuracy** across multi-subject behavioral therapy session videos.

- Processed, labeled, and annotated **50+ hours of video data**, enabling researchers to extract behavioral pattern insights **3× faster** than manual review workflows.
- Designed an efficient inference pipeline handling multi-person overlapping trajectories and occlusion scenarios, substantially improving model robustness in clinical settings.
- Collaborated directly with child psychology researchers to align technical output with clinical behavioral analysis frameworks.

Open Source Contributor

Apr 2024 – Jun 2025

AOSSIE (Australian Open Source Software Innovation and Education)

Remote

- Resolved a blocking TailwindCSS / PostCSS ESM configuration conflict that halted **all CI builds**, restoring a stable pipeline and unblocking **5+ active contributors**.
- Added full TypeScript support, GitHub Actions CI/CD workflows, and Docker configuration, **reducing manual deployment steps by ~60%** and eliminating environment-related onboarding failures.
- Improved developer experience and project documentation, accelerating contributor onboarding and code review cycles.

Freelance Full Stack & AI Developer

Apr 2025 – Present

Self-Employed

Remote

- Completed **10+ client projects** across AI, computer vision, data science, and full-stack web development, maintaining a **100% on-time delivery rate**.
- Published **3+ open-source ML and automation tools** on GitHub with active community adoption across the developer ecosystem.
- Delivered end-to-end solutions spanning requirements gathering, system design, implementation, testing, and deployment for diverse client verticals.

KEY PROJECTS

FAMS — AI Medical Assistant | *Python, Computer Vision, Voice AI, LLM Integration* 2025

- Architected a real-time visual assessment and voice-guided medical diagnosis pipeline integrating computer vision models with LLM-based reasoning for intelligent symptom analysis.
- Won **2nd Place at AVISHKAR 2025** (Bhilai, India) competing against **80+ teams**, demonstrating production-level AI integration under competitive conditions.

Money Muling Detector | *C++, React, TypeScript, Graph Algorithms, Decision Trees* 2025

- Designed an AI-powered financial fraud detection engine identifying money muling and smurfing patterns using a custom **Red-Black Tree + Decision Tree** hybrid algorithm at **O(N log N)** complexity.
- Processed and analyzed **10,000+ financial transactions** in real time; won **RIFT Hackathon** (Bangalore, India).

Computed Fields Engine | *TypeScript, Node.js, System Design, AsyncLocalStorage* 2025

- Built a generic Notion/Airtable-style computed fields system supporting formula-based field derivation (e.g., Price × Quantity, FullName concatenation) with dependency graph, topological sorting, and circular dependency detection.
- Resolved AsyncLocalStorage execution context issues to ensure safe server-side formula evaluation and consistent concurrent data updates.

SamarthyaBot — AI Agent Operating System | *Node.js, Multi-Agent AI, LLM Integration* 2025

- Built a privacy-first, self-hosted AI agent OS enabling autonomous workflows including code generation, browser automation, and deployment, with multi-LLM backend integration and secure local memory management.

FarmView AI — Smart Crop Monitoring | *YOLOv8, OpenCV, Machine Learning* 2025

- Created a real-time computer vision system capable of detecting **5+ categories** of crop disease, pest infestation, and nutrient deficiency directly from live camera input with actionable alerts for farmers.
- Presented as **Finalist at HackOmania 2025** (Bhilai, India) for innovation in agricultural AI.

Agentic HoneyPot — Cyber Threat Detection | *Python, AI Agents, Cybersecurity* 2025

- Designed an agent-based AI honeypot that simulates vulnerable network services to attract, log, and analyze attacker intrusion patterns, automatically generating structured threat intelligence reports.

Kuya Cloud — Data Analytics Platform | *Python, Data Processing, Visualization* 2025

- Created a rule-based analytics and automated visualization platform enabling non-technical users to explore datasets and generate insights with zero code, democratizing data science access.
- Published peer-reviewed research: *Sahu, B. P. (2026). Kuya Cloud. Zenodo. doi:10.5281/zenodo.18803300*

RESEARCH & PUBLICATIONS

Sahu, B. P. (2026). **Kuya Cloud: A Rule-Based Data Analytics and Automated Visualization Platform for Democratizing Data Science.** *Zenodo*. <https://doi.org/10.5281/zenodo.18803300>

EDUCATION

Bachelor of Computer Applications (BCA) <i>Rungta College of Science and Technology</i> Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Artificial Intelligence, Web Technologies.	Aug 2023 – Apr 2026 <i>Bhilai, India</i>
Generative AI Certification <i>NULL Class</i> Completed advanced training in Generative AI architectures, LLM fine-tuning, prompt engineering, and production deployment of AI systems.	Apr 2024 – Aug 2025 <i>Remote</i>

AWARDS & RECOGNITION

2nd Place — AVISHKAR 2025 <i>FAMS AI Medical Assistant Competed against 80+ teams</i>	Bhilai, India
Winner — RIFT Hackathon <i>Money Muling Detector / AI Financial Fraud Detection Engine</i>	Bangalore, India
Finalist — HackOmania 2025 <i>FarmView AI Smart Crop Monitoring System</i>	Bhilai, India
Participant — MumbaiHacks 2025 <i>FEZI Autonomous Fact-Verification System</i>	Mumbai, India